



ENGINEERING MATHEMATICS - II

UE20MA151

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Unit 4 : Inverse Laplace Transform

Session : 2

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- **First Shift or translation Theorem**
- **Change of Scale Property**
- **Inverse Laplace transform of Logarithmic and trigonometric functions**

First Shift or translation Theorem

If $L^{-1}\{F(s)\} = f(t)$ then,

$$L^{-1}\{F(s - a)\} = e^{at}f(t) = e^{at} L^{-1}\{F(s)\}$$

Recall !!!

$$L\{e^{at}f(t)\} = F(s - a)$$

Change of Scale Property

$$\text{If } L^{-1}\{F(s)\} = f(t) \text{ then } L^{-1}\{F(ks)\} = \frac{1}{k} f\left(\frac{t}{k}\right)$$

Recall !!!

$$L\{f(at)\} = \frac{1}{a} F\left\{\frac{s}{a}\right\}$$

Example : Find the inverse Laplace Transforms of $\frac{2s+3}{s^2+2s+2}$

Solution :
$$\begin{aligned} L^{-1} \left[\frac{2s+3}{s^2+2s+2} \right] &= L^{-1} \left[\frac{2s+3}{(s+1)^2+1} \right] \\ &= L^{-1} \left[\frac{2(s+1)+1}{(s+1)^2+1} \right] \\ &= 2L^{-1} \left[\frac{s+1}{(s+1)^2+1} \right] + L^{-1} \left[\frac{1}{(s+1)^2+1} \right] \\ &= 2e^{-t} L^{-1} \left[\frac{s}{s^2+1} \right] + e^{-t} L^{-1} \left[\frac{1}{s^2+1} \right] \\ &= 2e^{-t} \cos t + e^{-t} \sin t \end{aligned}$$

Example : If $L^{-1} \left\{ \frac{s^2-1}{(s^2+1)^2} \right\} = t \cos t$ then evaluate $L^{-1} \left\{ \frac{9s^2-1}{(9s^2+1)^2} \right\}$

Solution : Let $F(s) = \frac{s^2-1}{(s^2+1)^2}$.

Then, $L^{-1}[F(s)] = L^{-1} \left[\frac{s^2-1}{(s^2+1)^2} \right] = t \cos t = f(t)$.

Therefore, $L^{-1}[F(3s)] = L^{-1} \left[\frac{(3s)^2-1}{[(3s)^2+1]^2} \right] = \frac{1}{3} \cdot \frac{t}{3} \cos \frac{t}{3} = \frac{t}{9} \cos \frac{t}{3}$

Inverse Laplace transform of logarithmic and inverse trigonometric functions:

Given $F(s)$ we need to find $L^{-1}\{F(s)\} = f(t)$.

We have the property : $L\{t f(t)\} = -\frac{dL\{f(t)\}}{ds} = -F'(s)$.

Equivalently, $L^{-1}\{-F'(s)\} = t f(t) \dots \dots \dots (1)$

Working Rule:

- 1) In the case of Logarithmic functions we apply the properties of logarithms and then differentiate w.r.t s to obtain $F'(s)$.
- 2) Then multiply by -1 and take inverse on both sides.
- 3) LHS becomes $t f(t)$ by (1) and inverses are also found for the terms in RHS with the result we obtain the required $f(t)$.
- 4) If logarithmic function persists in $F'(s)$, we differentiate again w.r.t s to obtain $F''(s)$ and use the property that $L^{-1}\{F''(s)\} = t^2 f(t)$ since $L\{t^2 f(t)\} = F''(s)$.
- 5) In case of inverse functions we simply differentiate the given $F(s)$ and use the result (1) to obtain $f(t)$.

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INVERSE LAPLACE TRANSFORM



1) Find $L^{-1} \left\{ \log \left(\frac{s+a}{s+b} \right) \right\}$

Solution:

$$\text{Let } F(s) = \log \left(\frac{s+a}{s+b} \right) = \log(s+a) - \log(s+b)$$

$$F'(s) = \frac{1}{s+a} - \frac{1}{s+b}$$

$$-F'(s) = \frac{1}{s+b} - \frac{1}{s+a}$$

$$L^{-1}\{-F'(s)\} = L^{-1} \left(\frac{1}{s+b} \right) - L^{-1} \left(\frac{1}{s+a} \right)$$

$$tf(t) = e^{-bt} - e^{-at}$$

$$\text{Thus, } f(t) = \frac{e^{-bt} - e^{-at}}{t}$$



THANK YOU

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